

IMO(10th) – 1968

First Day

1. Prove that there exists a unique triangle whose side lengths are consecutive natural numbers and one of whose angles is twice the measure of one of the other.
2. Find all positive integer x for which $p(x) = x^2 - 10x - 22$, where $p(x)$ denotes the product of the digits of x .
3. Let a, b, c be real numbers. Prove that the system of equations:

$$\begin{cases} ax_1^2 + bx_1 + c = x_2 \\ ax_2^2 + bx_2 + c = x_3 \\ \dots\dots\dots \\ ax_{n-1}^2 + bx_{n-1} + c = x_n \\ ax_n^2 + bx_n + c = x_1 \end{cases}$$

- (a) has no real solution if $(b-1)^2 - 4ac < 0$;
- (b) has a unique real solution if $(b-1)^2 - 4ac = 0$;
- (c) has more than one real solution if $(b-1)^2 - 4ac > 0$;

Second Day

4. Prove that in any tetrahedron there is a vertex such that the lengths of its sides through that vertex are sides of a triangle.
5. Let $a > 0$ be a real number and $f(x)$ a real function defined on $\mathbb{R}$, satisfying for all $x \in \mathbb{R}$,

$$f(x+a) = \frac{1}{2} + \sqrt{f(x) - f(x)^2}.$$
 - (a) Prove that the function $f(x)$ is periodic, i.e. there exists $b > 0$ such that for all x ,

$$f(x+b) = f(x).$$
 - (b) Give an example of such a non-constant function for $a = 1$.
6. Let $[x]$ denote the integer part of x , i.e. the greatest integer not exceeding x . If n is a positive integer, express as a simple function of n the sum $\left[\frac{n+1}{2} \right] + \left[\frac{n+2}{4} \right] + \dots + \left[\frac{n+2^i}{2^{i+1}} \right] + \dots$

**